Chapter 11 Answers

Problem 1

An expression t is a term iff there is some sequence $t_1, ..., t_n$ such that every t_i is either a variable or a constant, or is an expression of the form $(t_j t_k)$ for some $j < i$ and $k < i$, and such that there is some sequence $k_1, ..., k_m$ such that every k_i is some t_i and $k_1 ... k_m = t$. Smullyan's answer seems to be incorrect here. For even consider the term $(x_1 x_2)$. According to his answer for Problem 1, we need to find a finite sequence $t_1 ... t_n = (x_1 x_2)$ such that for each $i \leq n$, either t_i is a variable or constant, or there are numbers j_1 and j_2 both less than i such that $t_i = (t_{j1} t_{j2})$. But such a sequence cannot be found. We would need to have some $t_i = (x_1 x_2)$. Because this is neither a constant nor a variable, we would need to have j_1 and j_2 such that $(x_1 x_2) = (t_{j1} t_{j2})$ for $j_1 < i$ and $j_2 < i$. Thus the sequence would be of the form $x_1 x_2 (x_1 x_2)$. But $x_1 x_2 (x_1 x_2) \neq (x_1 x_2)$. Thus this definition does not work. Instead, with the preceeding definition, one could construct the sequence $t_1 t_2 t_3 = x_1 x_2 (x_1 x_2)$ and from that construct the sequence $k_1 = (x_1 x_2)$, which would show that $(x_1 x_2)$ is a term.

Justification for Term Induction

Here is why term induction can be derived from mathematical induction. Now, suppose S is a set whose members are variables or constants or both. Let S^+ be the set of all terms built from elements of S (and parantheses). Now suppose that a property P holds for all elements of S, and if any terms t_1 and t_2 have property P, then so does $(t_1 t_2)$. Now, S^+ consists of all variables and constants in S, in fact, all elements of S. We wish to show that for all n being the number of occurrences of elements of S in a term in S^+, that term has property P. The base case is obvious, as terms having only 1 occurence are either variables or constants, and these are assumed to have property P (as they are all in S). Now suppose the assumption holds for all number $< n$. A term having n occurences of elements of S must be of the form $(t_1 t_2)$ where t_1, t_2 both have less than n occurences of elements of S in them. Hence both t_1 and t_2 have property P, meaning so does $(t_1 t_2)$, that is, t. Thus we have proven by complete induction that all elements of S^+ have the property P.

Problem 2

(i): Suppose that $\mathcal{W}$ is x. Hence $\mathcal{W}_x$ is I. (1) and (2) hold, as I contains no variables. Also, $\mathcal{W}_x x = Ix = x = \mathcal{W}$, meaning condition (3) holds. Hence I is an x-eliminate of $\mathcal{W}$.

(ii): Suppose that $\mathcal{W}$ is any term in which x does not occur. Hence $\mathcal{W}_x$ is $K\mathcal{W}$. (1) holds, as $K\mathcal{W}$ contains only those variables in $\mathcal{W}$, and $\mathcal{W}$ does not contain x; this also means (2) holds. (3) also holds, as $K\mathcal{W}x = \mathcal{W}$. Thus $K\mathcal{W}$ is an x-eliminate of $\mathcal{W}$.

(iii): Suppose that $\mathcal{W}_x$ and $\mathcal{V}_x$ are x-eliminates of $\mathcal{W}$ and $\mathcal{V}$, respectively. Thus $(\mathcal{WV})_x$ is $S\mathcal{W}_x\mathcal{V}_x$. (1) holds, as x was assumed to not occur in either of $\mathcal{W}_x$ or $\mathcal{V}_x$. (2) holds in the following way: The variables of $S\mathcal{W}_x\mathcal{V}_x$ are those of $\mathcal{W}$ and $\mathcal{V}$. The variables in $(\mathcal{WV})$ are those in $\mathcal{W}$ or in $\mathcal{V}$. Because we assumed the variables of $\mathcal{W}_x$ and $\mathcal{V}_x$ occured in $\mathcal{W}$ and $\mathcal{V}$, respectively, it thus follows that the variables of $S\mathcal{W}_x\mathcal{V}_x$ are all in $(\mathcal{WV})$. (3) also holds, as $S\mathcal{W}_x\mathcal{V}_x x = \mathcal{W}_x x(\mathcal{V}x) = \mathcal{W}(\mathcal{V}) = \mathcal{WV}$. Thus $S\mathcal{W}_x\mathcal{V}_x$ is an x-eliminate of $(\mathcal{WV})$.

Note that this is why Proposition 1 follows from Problem 2 by term induction. We wish S^+ to be the set of all simple terms. Thus let S be the set of all variables, as well as the constants S, K and I. Hence S^+ is the set of all terms built from elements of S and parentheses, that is, all simple terms. Let P be the property that $\mathcal{W}_x$ is an x-eliminate of $\mathcal{W}$. We have shown by (i) that $\mathcal{W}_x$ for $\mathcal{W}$ being a variable is an x-eliminate of $\mathcal{W}$. Similarly, for S, K and I, we showed in (ii) that their respective $\mathcal{W}_x$'s are x-eliminates of them. Thus we have shown that P holds for all elements of S. Now suppose we have any terms t_1 and t_2 of S^+ such that t_1 and t_2 have the property P. Because t_1 and t_2 are simple terms, we thus showed in (iii) that (t_1t_2) also has property P. Thus, by term induction, P holds for all elements in S^+, which is what we were required to prove.

Problem 3

Suppose that x does not occur in $\mathcal{W}$. We prove that $\mathcal{W}$ is an x-eliminate of $\mathcal{W}x$. (1) holds, as x was assumed to not occur in $\mathcal{W}$. (2) obviously holds, as $\mathcal{W}$ is itself part of $\mathcal{W}x$. (3) also obviously holds, as $\mathcal{W}_x x = \mathcal{W}x = \mathcal{W}x$.

Problem 4

(2) is the one which is true. For consider $\mathcal{W}_{x,y}$. (1) and (2) obviously hold. Because $\mathcal{W}_{x,y} = (\mathcal{W}_x)_y$, we have $\mathcal{W}_{x,y}yx = (\mathcal{W}_x)_y yx = \mathcal{W}_x x = \mathcal{W}$, meaning (3) also holds.

Here is an explicit proof of Proposition 2 from Problem 4 and mathematical induction. The base cases of Proposition 2, namely when $n \leq 2$, are just Problems 2 and 4. Now suppose the result holds for some n, that is, $\mathcal{W}_{x_n x_{n-1} \ldots x_1}$ is an $x_1, x_2, ..., x_n$ eliminate of $\mathcal{W}$. Now, $\mathcal{W}_{x_{n+1}x_n \ldots x_1}$ is $(...(\mathcal{W}_{x_{n+1}})_{x_n})...)_{x_1})$. Conditions (1) and (2) obviously hold. As for (3),

$$\begin{aligned}
\mathcal{W}_{x_{n+1}x_n \ldots x_1} x_1 x_2 ... x_{n+1} &= (...(\mathcal{W}_{x_{n+1}})_{x_n})...)_{x_1}) x_1 x_2 ... x_{n+1} \\
&= (...(\mathcal{W}_{x_{n+1}})_{x_n})...)_{x_2}) x_2 ... x_{n+1} \\
&= \mathcal{W}_{x_{n+1}x_n \ldots x_2} x_2 ... x_{n+1} \\
&= \mathcal{W}
\end{aligned}$$

by the induction assumption.

Problem 5

(1): We can prove this in two ways. First, suppose that A is the combinator which realizes the term $t(x, x_1, ..., x_n)$. Because we are working in a complete system, we can consider the fixed point A' of A. Take Γ to be AA' (or, equivalently, just A'), as $AA'x_1...x_n = t(A', x_1, ..., x_n) = t(AA'(x_1, ..., x_n))$. An alternative proof instead supposes that A is the combinator which realizes the term $t((xx), x_1, ..., x_n)$. In this case, we take Γ to be AA, as $AAx_1, ..., x_n = t(AA, x_1, ..., x_n)$.

Note to self: Someone could ask: "We have already proved that there is some combinator that realizes every simple term t. So, if we have a given simple term $t(x, x_1, ..., x_n)$, we could, for *any* combinator A, take t_1 to be $t(A, x_1, ..., x_n)$, and then there is some combinator B which realizes t_1. Doesn't this prove the fixed point theorem?" It shows that there is, for each combinator A, *some other* combinator B that realizes $t(A, x_1, ..., x_n)$. But this means B could be something other than A. The theorem wants to have A and B be the same, so this line of reasoning does not work.

(2): Part (a) of the exercise is a special case when $t(x_1, x, y)$ is the term $x(x_1y)$ (this is just the queer bird Q, as $Qxyz = y(xz)$), as then $\Gamma xy = x(\Gamma y)$. Part (b) of the exercise follows when $t(x_1, x, y)$ is the term x_1yx (this is just the cardinal C, which was defined as $Cxyz = xzy$), as then $\Gamma xy = \Gamma yx$.

Problem 6

Suppose the terms $t_1(x, x_1, ..., x_n)$ and $t_2(x, x_1, ..., x_n)$ are realizable by A_1 and A_2, respectively. Because the system is complete, we have the combinator $A_1(A_2(BA_1A_2)')$ for $(BA_1A_2)'$ the fixed point of (BA_1A_2). Take Γ_1 to be $A_1(A_2(BA_1A_2)')$, and Γ_2 to be $A_2(BA_1A_2)'$. Then:

$$A_1(A_2(BA_1A_2)')x_1...x_n = t_1(A_2(BA_1A_2)', x_1, ..., x_n)$$

and

$$\begin{aligned} A_2(BA_1A_2)'x_1....x_n &= t_2((BA_1A_2)', x_1, ..., x_n) \\ &= t_2(BA_1A_2(BA_1A_2)', x_1, ..., x_n) \\ &= t_2(A_1(A_2(BA_1A_2)'), x_1, ..., x_n) \end{aligned}$$

That is:

$$\begin{aligned} \Gamma_1x_1...x_n &= t_1(\Gamma_2, x_1, ..., x_n) \\ \Gamma_2x_1...x_n &= t_2(\Gamma_1, x_1, ..., x_n) \end{aligned}$$

Smullyan provides two alternative solutions. One takes A_1 and A_2 to be the combinators which realize $t_1(xyx, x_1, ..., x_n)$ and $t_2(xyx, x_1, ..., x_n)$, respectively. That is:

$$A_1xyx_1...x_n = t_1(xyx, x_1, ..., x_n)$$

$$A_2xyx_1...x_n = t_2(xyx, x_1, ..., x_n)$$

and then the theorem is satisfied by taking Γ_1 to be $A_1A_2A_1$ and Γ_2 to be $A_2A_1A_2$, as:

$$A_1A_2A_1 = t_1(A_2A_1A_2, x_1, ..., x_n)$$

$$A_2A_1A_2 = t_2(A_1A_2A_1, x_1, ..., x_n)$$

The other proof uses the previously introduced notion of a cross point (x, y) of a pair (A, B) meaning $Ax = y$ and $By = x$. By Problem 6 of Chapter 8, and the fact that we are working in a complete system, every pair of terms has a cross point. Thus, if A_1 and A_2 realize t_1 and t_2 respectively, we let (Γ_1, Γ_2) be the cross point of (A_2, A_1), as then $A_2\Gamma_1 = \Gamma_2$ and $A_1\Gamma_2 = \Gamma_1$, meaning:

$$\Gamma_1x_1...x_n = A_1\Gamma_2x_1...x_n = t_1(\Gamma_2, x_1, ..., x_n)$$

$$\Gamma_2x_1...x_n = A_2\Gamma_1x_1...x_n = t_2(\Gamma_1, x_1, ..., x_n)$$

Problem 7

We wish to prove Proposition 3. This can be done by using term induction. Take the set S to be all of the variables, as well as S, B, C and I (the only constants in our system). Hence S^+ is the set of all nice terms. Now, let P be the property that, if the term $\mathcal{W}$ is nice and such that x occurs in it, then $\mathcal{W}_x$ is a nice-x-eliminate of $\mathcal{W}$. We need to first show that P holds for all elements of S, and then show that it holds for all elements of S^+. Well, for every element of S, it either is x or it isn't. If it isn't, the property vacuosly holds. If it is x, then $\mathcal{W}_x$ is I. Recall the conditions of being a nice-x-eliminate of $\mathcal{W}$:
(1) The variable x does not occur in $\mathcal{W}_x$.
(2) All variables of $\mathcal{W}_x$ occur in $\mathcal{W}$.
(3) The equation $\mathcal{W}_x x = \mathcal{W}$ holds.
Because I has no variables, conditions (1) and (2) obviously hold. (3) also holds, as $Ix = x = \mathcal{W}$. Thus I is an nice-x-eliminate of x. We have therefore shown that P holds for all elements in S. Now consider any terms t_1 and t_2 of S^+. Suppose t_1 and t_2 have property P. Consider the term $(t_1 t_2)$. There are now multiple cases.

In the first case, x occurs both in t_1 and t_2. Thus $(t_1 t_2)_x$ is $S(t_1)_x(t_2)_x$. Conditions (1) and (2) obviously hold (by the assumption of t_1 and t_2 having property P). (3) also holds, as $S(t_1)_x(t_2)_x x = (t_1)_x x((t_2)_x x) = t_1(t_2) = t_1 t_2$.

In the second case, x occurs in t_1 but not in t_2. Thus $(t_1 t_2)_x$ is $C(t_1)_x t_2$. Conditions (1) and (2) obviously hold. (3) also holds, as $C(t_1)_x t_2 x = (t_1)_x x t_2 = t_1 t_2$.

In the third case, x occurs in t_2 but not in t_1. There are then two sub-cases. In the first, t_2 is just x. Then $(t_1 t_2)_x$ is t_1. Again, conditions (1) and (2) obviously hold. As for (3), it also holds, as $t_1 x = t_1 t_2$ (as we assumed t_2 was just x). In the second sub-case, $t_2 \neq x$, but x occurs in t_2 but not in t_1. Then $(t_1 t_2)_x$ is $Bt_1(t_2)_x$. Again, (1) and (2) obviously hold. As for (3), it also holds, as $Bt_1(t_2)_x x = t_1((t_2)_x x) = t_1(t_2) = t_1 t_2$.

The last case is the vacuous one, in which neither t_1 or t_2 contain x. Then $(t_1 t_2)$ would also not contain x, meaning P would hold vacuously for it.

Thus, in all cases, $(t_1 t_2)$ also has the property P. Then, by term induction, all elements of S^+ have the property P. As a consequence of this, for any nice term $\mathcal{W}$ in which x occurs, $\mathcal{W}_x$ is a nice-x-eliminate of $\mathcal{W}$.

Here is the proof of the generalization of this result: Suppose that the variables $x_1, ..., x_n$ occur in some nice term $\mathcal{W}$. We then must show that $\mathcal{W}_{x_n x_{n-1} ... x_1}$ is a nice $x_1, ..., x_n$ eliminate of $\mathcal{W}$. Recall that $\mathcal{W}_{x_n x_{n-1} ... x_1}$ is an abbreviation for $(...(\mathcal{W}_{x_n})_{x_{n-1}})...)_{x_1})$. This can be proven by induction on n. The base case is just the above result, when we showed that for any nice term $\mathcal{W}$ in which x occurs, the term $\mathcal{W}_x$ is a nice x-eliminate of $\mathcal{W}$. Now suppose the result holds for some n. Now, conditions (1) and (2) obviously hold for $\mathcal{W}_{x_{n+1} x_n ... x_1}$. As for (3),

$$\mathcal{W}_{x_{n+1} x_n ... x_1} x_1 x_2 ... x_{n+1} = (...(\mathcal{W}_{x_{n+1}})_{x_n})...)_{x_1}) x_1 x_2 ... x_{n+1}$$

$$= (...(\mathcal{W}_{x_{n+1}})_{x_n})...)_{x_2}) x_2 ... x_{n+1}$$

$$= \mathcal{W}_{x_{n+1} x_n ... x_2} x_2 ... x_{n+1}$$

$$= \mathcal{W}$$

by the induction assumption.

Problem 8

(a): Take Q_3 to be GI, as $GIxyz = Iz(xy) = z(xy)$.

(b): Take C to be $GGII$, as $GGIIxyz = Gx(II)yz = xz(IIy) = xzy$.

(c): Take T to be CI, as $CIxy = Iyx = yx$. Alternatively, we could take T to be Q_3I, as $Q_3Ixy = y(Ix) = y(x) = yx$.

(d): Recall that we can take R to be CC, as $CCxyz = Cyxz = yzx$. Then take Q to be $C(Q_3Q_3)C$, as $C(Q_3Q_3)Cxyz = Q_3Q_3xCyz = C(Q_3x)yz = Q_3xzy = y(xz)$. Alternatively, we could take Q to be $RC(Q_3Q_3)$, as $RC(Q_3Q_3)xyz = Q_3Q_3xCyz = C(Q_3x)yz = Q_3xzy = y(xz)$. Yet another alternative solution is to take Q to be GRQ_3, as $GRQ_3xyz = Ry(Q_3x)z = Q_3xzy = y(xz)$. Then, take B to be CQ, as $CQxyz = Qyxz = x(yz)$.

Derivations of and from J and I

Here are derivations for B, T and M from J and I. We can easily see that JII is T, as:

$$JIIxy = II(Iyx) = Iyx = yx$$

Then, we can take R to be JT, as $JTxyz = Tx(Tzy) = Tzyx = yzx$, and we can thus take C to be RRR, that is, $JT(JT)(JT)$. We then take Q_2 to be $C(JI)$, as $C(JI)xyz = JIyxz = Iy(Izx) = y(zx)$. Then it follows that Q_1 is also present, as $CQ_2xyz = Q_2yxz = x(zy)$. Then we also have Q_4, as $Q_1Txyz = T(yx)z = z(yx)$. Then we also have Q_3, in particular, from CQ_4, as $CQ_4xyz = Q_4yxz = z(xy)$ (the derivations for Q_3 and Q_4 are not necessary).

Next, we can derive B. Take B to be $C(JIC)Q_1$, as

$$C(JIC)Q_1xyz = JICxQ_1yz = IC(IQ_1x)yz = C(Q_1x)yz = Q_1xzy = x(yz)$$

Now having derived B and T, it follows from problems in Chapter 9 that we have the vireo V (which we recall was defined by the condition $Vxyz = zxy$), and also V^*. We can then take our mocker M to be $V^*(B(CJI)T)IT$, as

$$V^*(B(CJI)T)ITx = B(CJI)TxIT = CJI(Tx)IT = J(Tx)IIT = TxI(TxTI)$$

$$= Ix(TxI) = x(Ix) = x(x) = xx$$

Thus, we have derived T, B and M (in that order) from J and I.

In *To Mock a Mockingbird*, Smullyan provides alternative derivations for B and M. He first derives Q_1, T and R all in the same manner as above. Then he takes C^* to be $C(Q_1C)$, as $C(Q_1C)xyzw = Q_1Cyxzw = C(xy)zw = xywz$. After that, he takes B to be C^*Q_1, as $C^*Q_1xyz = Q_1xzy = x(yz)$. He then defines a new bird J_1 defined by $J_1xyzw = yx(wxz)$, which he takes to be BJT, as $BJTxyzw = J(Tx)yzw = Txy(Txwz) = yx(wxz)$. Then he notes that $J_1xTTT = Tx(TxT) = TxTx = Txx = xx$. In turn, $J_1xTTT = CJ_1TxTT = C(CJ_1T)TxT = C(C(CJ_1T)T)Tx$. Hence, he takes M to be $C(C(CJ_1T)T)T$.

It is also true that we can derive J from B, T and M. Because we have B, T and M, we have, by problems in Chapter 9, W, C^{***}, C^*, and E. Then take J to be $W(C^{***}(C^*E))$, as

$$W(C^{***}(C^*E))xyzw = C^{***}(C^*E)xxyzw = C^*Exxywz = Exyxwz = xy(xwz)$$

In *To Mock a Mockingbird*, Smullyan provides a slightly different derivation of J. He takes it to be $C^{**}(W(C^*E))$, as $C^{**}(W(C^*E))xyzw = W(C^*E)xywz = C^*Exxywz = Exyxwz = xy(xwz)$.